

Question 1. How many surgeries did each patient have? Your results should show patient id, patient names as first and last name separated by space, and number of surgeries ordered by the total number of surgeries in descending order. Note, patients who did not have surgeries must not be listed.

SELECT

```
'1729171' AS StuID,  
p.patient_id AS pID,  
CONCAT(p.f_name, ' ', p.l_name) AS pName,  
COUNT(*) AS sCount
```

FROM patients9171 p

INNER JOIN surgeryrecord9171 sr ON p.patient_id = sr.patient_id

GROUP BY pID

ORDER BY sCount DESC;

	StuID	pID	pName	sCount
▶	1729171	494	Waleed Kazmi	5
	1729171	507	Usman Mirza	5
	1729171	216	Shaista Yasin	4
	1729171	605	Rayyan Ali	4
	1729171	1114	Jibran Younus	4
	1729171	1582	Iram Qadir	4
	1729171	132	Ammar Aziz	3
	1729171	173	Ifrah Mirza	3
	1729171	213	Huda Kazmi	3
	1729171	377	Rashid Bhatti	3

734 rows returned

Question 2. We need a list of doctors, showing for each doctor the total amount of payments for the doctor's appointments with patients. The list should also include doctors who did not have any appointments. Your results should show doctor IDs, doctor names as first and last names separated by space, and total amounts, ordered by the totals from highest to lowest.

SELECT

'1729171' AS StuID,

d.doct_id AS dID,

CONCAT(d.f_name, ' ', d.l_name) AS dName,

IFNULL(SUM(a.payment_amount), 0) AS dTotalAmountEarned

FROM doctor9171 AS d

LEFT JOIN appointment9171 a ON d.doct_id = a.doct_id

GROUP BY dID

ORDER BY dTotalAmountEarned DESC;

	StuID	dID	dName	dTotalAmountEarned
	1729171	1027	Dr. Raza Hasnain	830436
	1729171	1020	Dr. Waqas Javed	184568
	1729171	1024	Dr. Muhammad Imran	183015
	1729171	1001	Dr. Najeeb Basir	134005
	1729171	1008	Dr. Tahir Hamid	99899
	1729171	1009	Dr. Muhammad Saleem Ahmad Siddiqui	95574
	1729171	1002	Dr. Kamran Ahmed	50356
	1729171	1004	Dr. Hanif Kamal	44867
	1729171	1003	Dr. Sheikh M. Ilyas	40634
	1729171	1005	Dr. Muhammad Haroon Hamid	0

400 rows returned

Question 3. Produce a report showing number of shifts only with doctors on duty (no nurses or helpers), number of shifts with only nurses on duty (no doctors or helpers) and number of shifts with only helpers on duty (no doctors or nurses). Your report should use the following format and headings (actual staff numbers are removed):

StuID	Staff on duty	Total shifts
123459999	doctor shift	114
123459999	nurse shift	51
123459999	helper shift	221

SELECT

'1729171' AS StuID,

'doctor shift' AS `Staff on duty` ,

SUM(doct_id IS NOT NULL AND nurse_id IS NULL AND helper_id IS NULL) AS `Total shifts`

FROM staffshift9171

UNION ALL

SELECT

'1729171' AS StuID,

'nurse shift' AS `Staff on duty` ,

SUM(doct_id IS NULL AND nurse_id IS NOT NULL AND helper_id IS NULL) AS `Total shifts`

FROM staffshift9171

UNION ALL

SELECT

'1729171' AS StuID,

'helper shift' AS `Staff on duty` ,

SUM(doct_id IS NULL AND nurse_id IS NULL AND helper_id IS NOT NULL) AS `Total shifts`

FROM staffshift9171;

	StuID	Staff on duty	Total shifts
▶	1729171	doctor shift	1058
	1729171	nurse shift	500
	1729171	helper shift	500

3 rows returned

Question 4. List all patients who have more than one cancelled appointment. Your list should show patient id, last name and first name, gender and how many appointments this patient cancelled. Order the results by patients last name.

SELECT

'1729171' AS StuID,
p.patient_id AS pID,
p.l_name AS lName,
p.f_name AS fName,
p.gender,
a.appointment_status AS aStatus,

COUNT(*) AS numCancelledAppts

FROM patients9171 AS p

LEFT JOIN appointment9171 a ON p.patient_id = a.patient_id

WHERE a.appointment_status = 'Cancelled'

GROUP BY p.patient_id

HAVING COUNT(*) > 1

ORDER BY lName;

	StuID	pID	lName	fName	gender	aStatus	numCancelledAppts
►	1729171	1168	Iqbal	Marium	F	Cancelled	2

1 row returned

Question 5. List all helpers who have not been allocated to any staff shifts. Your list should show department id and name where the helper works, helper id and helper full name as first and last names separated by space. The results should be sorted by department id and within each department alphabetically by helper full name.

SELECT

```
'1729171' AS StuID,
d.dept_id AS deptID,
d.dept_name AS deptName,
h.helper_id AS hID,
CONCAT(h.f_name, ' ', h.l_name) AS hName
```

FROM helpers9171 h

LEFT JOIN staffshift9171 ss ON h.helper_id = ss.helper_id

LEFT JOIN department9171 d ON h.dept_id = d.dept_id

WHERE ss.shift_id IS NULL

ORDER BY deptID, hName;

	StuID	deptID	deptName	hID	hName
▶	1729171	101	Cardiology	8816	Abrar Butt
	1729171	101	Cardiology	8540	Ahmed Warraich
	1729171	101	Cardiology	8765	Dania Awan
	1729171	101	Cardiology	8530	Durr-e-Shehwar Jafri
	1729171	101	Cardiology	8954	Erum Zaidi
	1729171	101	Cardiology	8878	Ghazala Awan
	1729171	101	Cardiology	9011	Kashif Bukhari
	1729171	101	Cardiology	8896	Kishwar Tabassum
	1729171	101	Cardiology	8725	Naveed Naqvi
	1729171	101	Cardiology	8756	Parveen Raja

600 rows returned

Question 6. How much money came through as payments in total and in each calendar year using each mode of payment. Your results should show full total in the first row and then rows with details: calendar year, mode of payment and money received ordered by year and mode of payment (similar to below but showing total values in the last column). Your output should look similar to the one below.

StuID	CalendarYear	mode_of_payment	PaymentsTotal
123459999		Total	-----
123459999	2024	Card	-----
123459999	2024	Cash	-----

SELECT

 '1729171' AS StuID,
 YEAR(appointment_date) AS calendarYear,
 mode_of_payment AS modeOfPayment,
 SUM(payment_amount) AS paymentsTotal,
 1 AS rowType,
 YEAR(appointment_date) AS sortYear

FROM appointment9171

GROUP BY YEAR(appointment_date), mode_of_payment

UNION ALL

SELECT

 '1729171' AS StuID,
 " AS calendarYear,
 'Total' AS modeOfPayment,
 SUM(payment_amount) AS paymentsTotal,
 0 AS rowType,
 YEAR(appointment_date) AS sortYear

FROM appointment9171

GROUP BY YEAR(appointment_date)

ORDER BY sortYear ASC, rowType ASC, modeOfPayment ASC;

	StuID	calendarYear	modeOfPayment	paymentsTotal	rowType	sortYear
►	1729171		Total	813144	0	2024
	1729171	2024	Card	222744	1	2024
	1729171	2024	Cash	182143	1	2024
	1729171	2024	Digital Wallet	221040	1	2024
	1729171	2024	Insurance	187217	1	2024
	1729171		Total	850210	0	2025
	1729171	2025	Card	179871	1	2025
	1729171	2025	Cash	226441	1	2025
	1729171	2025	Digital Wallet	217424	1	2025
	1729171	2025	Insurance	226474	1	2025

10 rows returned

Question 7. List the number of surgeries performed by each surgeon in each calendar year. Your list should display years when surgeries were performed, then surgeons' id and last name and the number of performed surgeries.

SELECT

'1729171' AS StuID,

d.doct_id AS sID,

d.l_name AS sLastName,

IFNULL(YEAR(sr.surgery_date), '') AS yearOfSurgery,

IFNULL(COUNT(surgery_id), 0) AS numSurgeries

FROM doctor9171 d

LEFT JOIN surgeryrecord9171 sr

ON d.doct_id = sr.surgeon_id

GROUP BY IFNULL(YEAR(sr.surgery_date), ''), d.doct_id;

	StuID	sID	sLastName	yearOfSurgery	numSurgeries
►	1729171	1001	Basir		0
	1729171	1002	Ahmed		0
	1729171	1003	M. Ilyas		0
	1729171	1004	Kamal		0
	1729171	1005	Haroon Hamid		0
	1729171	1006	Farooq		0
	1729171	1007	Najeeb		0
	1729171	1008	Hamid		0
	1729171	1009	Saleem Ahmad Siddiqui		0
	1729171	1010	Bhatti		0

509 rows returned

Question 8. For each letter of the alphabet – A, B, C etc. – display the number of patients whose last name begins with that letter. Order your output alphabetically. If a letter doesn't have any corresponding patients, you can simply omit it from the listing. Hint: you can use the LEFT(string, n) function to find the first letter of a string. For example, the first two lines of your output would look like this:

Letter	HowMany
A	241
B	71

SELECT

'1729171' AS StuID,

LEFT(p.l_name, 1) AS Letter,

COUNT(*) AS HowMany

FROM patients9171 p

GROUP BY Letter

ORDER BY Letter;

	StuID	Letter	HowMany
▶	1729171	A	241
	1729171	B	71
	1729171	D	41
	1729171	F	38
	1729171	G	37
	1729171	H	91
	1729171	I	47
	1729171	J	72
	1729171	K	116
	1729171	L	54

19 rows returned

Question 9. Which patient (or patients if more than one) has had the highest number of appointments? (That is, more appointments than all the other patients.) If there is a tie for first place, list all the patients who tied. Your results should show the patient(s)' ID, first and last name in the format "last comma space first" (e.g. Lee, June), and the number of appointments they had. Hint: you will need to use ALL (see additional explanation https://www.w3schools.com/sql/sql_all.asp).

SELECT

```
'1729171' AS StuID,  
p.patient_id AS pID,  
CONCAT(L_name, ', ', f_name) AS pName,  
COUNT(*) AS numAppts
```

FROM patients9171 p

INNER JOIN appointment9171 a

ON p.patient_id = a.patient_id

GROUP BY p.patient_id

HAVING COUNT(*) >= ALL(

```
SELECT COUNT(*)
```

```
FROM appointment9171 a2
```

```
GROUP BY a2.patient_id
```

);

	StuID	pID	pName	numAppts
▶	1729171	527	Shuja, Ameena	5
	1729171	1455	Hashmi, Eman	5

2 rows returned

Question 10a. Add yourself as a patient. You do not need to provide your real DOB, phone and address, any fictitious data is acceptable, as long as it is saved in the table. Use your student ID as patient ID. Provide a screenshot of the message generated by Workbench that a record was added.

```
INSERT INTO patients9171
```

```
VALUES(1729171, 'Timothy', 'Wong', 'M', '1999-12-25', '0467897622', '1 Park Road,  
Carlton VIC 3010, Australia');
```

```
✓ 16 13:50:25 INSERT INTO patients9171 VALUES(1729171, 'Timothy', 'Wong', 'M', '1999-12-25', '0467897622', '1 Park Road, Carlton VIC 3010, Australia') 1 row(s) affected 0.000 sec
```

Question 10b. Add an appointment for yourself with Dr. Sonia Ali on any date in October 2026; use any reason you want for the appointment. The appointment status should be as “scheduled” and in person without payment details. This task needs to be done as a simple transaction where you need to determine doctor’s ID first and only after that you can save appointment details.

Note, to find out the last appointment number use MAX(appointment_id). This task needs to be done as a transaction.

Provide a screenshot of the status of executing the transaction. Make sure you include all status rows relevant to the transaction.

START TRANSACTION;

-- Step 1. Find Dr. Sonia Ali's ID

```
SET @DrSoniaAliID = (
    SELECT doct_id AS `Dr. Sonia Ali's ID`
    FROM doctor9171
    WHERE l_name = 'Ali' AND f_name = 'Dr. Sonia'
);
```

-- Step 2. Find the last appointment number

```
SET @LastApptNum = (
    SELECT MAX(appointment_id) AS `Last appointment number`
    FROM appointment9171
);
```

-- Step 3. Schedule an appointment

```
INSERT INTO appointment9171(appointment_id, patient_id, doct_id, appointment_date,
appointment_status)
```

```
VALUES(@LastApptNum + 1, 1729171, @DrSoniaAliID, '2026-10-15', 'Scheduled');
```

COMMIT;

22	14:14:18	START TRANSACTION	0 row(s) affected	0.016 sec
23	14:14:22	SET @DrSoniaAliID = (SELECT doct_id AS 'Dr. Sonia Ali's ID' FROM doctor9171 WHERE l_name = 'Ali' AND f_name = 'Dr. Sonia')	0 row(s) affected	0.000 sec
24	14:14:25	SET @LastApptNum = (SELECT MAX(appointment_id) AS 'Last appointment number' FROM appointment9171)	0 row(s) affected	0.016 sec
25	14:14:27	INSERT INTO appointment9171(appointment_id, patient_id, doct_id, appointment_date, appointment_status) VALUES(@LastApptNum + 1, 1729171, @DrSoniaAliID, '2026-10-15', 'Scheduled')	1 row(s) affected	0.015 sec
26	14:14:29	COMMIT	0 row(s) affected	0.000 sec

Question 10c. Create an SQL statement showing your id and name as a patient and your appointment details including appointment id, doctor's id and full name (last and first separated by comma and space), date, mode of the appointment and status of the appointment.

SELECT

```
'1729171' AS StuID,  
p.patient_id AS pID,  
CONCAT(p.l_name, ', ', p.f_name) AS pName,  
appointment_id AS apptID,  
d.doct_id AS dID,  
CONCAT(d.l_name, ', ', d.f_name) AS dName,  
appointment_date AS apptDate,  
mode_of_appointment AS apptMode,  
mode_of_payment AS paymentMode,  
appointment_status AS apptStatus
```

FROM patients9171 p

LEFT JOIN appointment9171 a ON p.patient_id = a.patient_id

LEFT JOIN doctor9171 d ON d.doct_id = a.doct_id

WHERE p.patient_id = 1729171;

	StuID	pID	pName	apptID	dID	dName	apptDate	apptMode	paymentMode	apptStatus
►	1729171	1729171	Wong, Timothy	6000	1018	Ali, Dr. Sonia	2026-10-15	NULL	NULL	Scheduled

1 row returned

Question 11a. Write the SQL DDL to create a view that counts number of wards in each department. Your list should include department ID, department name and number of wards.

CREATE VIEW WardCount AS

SELECT

d.dept_id AS deptID,

d.dept_name AS deptName,

COUNT(*) AS numWards

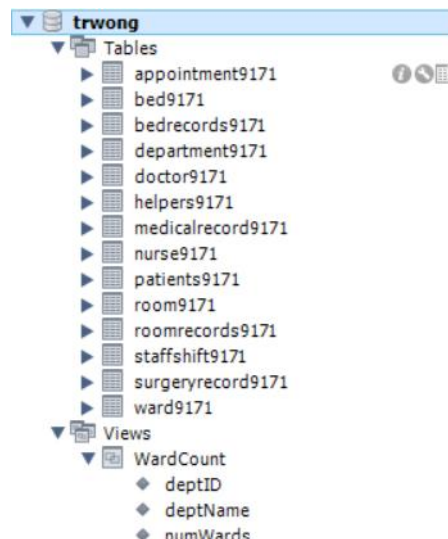
FROM department9171 d

LEFT JOIN ward9171 w ON d.dept_id = w.dept_id

GROUP BY d.dept_id;

In addition to the code, you need to provide 2 screenshots:

1. the list of tables and views from the left pane of Workbench showing your created view



2. the results of running SELECT from your View (we recommend explicitly showing the SELECT statement used to create the View).

	deptID	deptName	numWards
▶	101	Cardiology	5
	102	Neurology	4
	103	Gastroenterology	1
	104	Nephrology	2
	105	Pulmonology	2
	106	Endocrinology	1
	107	General Medicine	5
	108	Pediatrics	3
	109	Orthopedics	2
	110	Gynecology	3

31 rows returned

Question 11b. Using the View you created in Task 11a, list the departments with the highest number of wards. Your results need to display department id, department name and the corresponding number of wards. You must include your student ID in task b.

SELECT

'1729171' AS StuID,

WardCount.*

FROM WardCount

WHERE numWards = (

SELECT MAX(numWards)

FROM WardCount

);

	StuID	deptID	deptName	numWards
►	1729171	101	Cardiology	5
	1729171	107	General Medicine	5

2 rows returned

